

Additional Exercises 3

Linear Approximation, Total Differential & Chain Rule

Mirrors: List 4 (exercises 3–9) | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Find the total differential

- (a) ★ $z = x^3y - y^3x$
- (b) ★ $z = e^x \sin(xy)$
- (c) ★ $u = x^2 + y^2 + z^2$
- (d) ★ $z = x^2y^3$
- (e) ★★ $z = \arctan\left(\frac{y}{x}\right)$
- (f) ★★ $u = xyz + x^2z$
- (g) ★★ $z = \ln\left(\frac{x+y}{x-y}\right)$
- (h) ★★ $u = \sqrt{x^2 + y^2 + z^2}$
- (i) ★★★ $z = x^y$
- (j) ★★★ $u = f(x^2y, y^2z), f$ differentiable.

Exercise 2: Approximate using the total differential

Find a nearby “nice” point (a, b) , then use $f(a + \Delta x, b + \Delta y) \approx f(a, b) + f_x \Delta x + f_y \Delta y$.

- (a) ★ $1.02^{4.05}$
- (b) ★ $\ln(0.09^2 + 0.99^2)$
- (c) ★ $\arctan\left(\frac{1.05}{0.98}\right)$
- (d) ★★ $1.02^3 \cdot 0.97^2$
- (e) ★★ $\sqrt{5e^{0.02} + 2.03^2}$
- (f) ★★★ $\sqrt[4]{4.05^2 + 2.93^2}$

Exercise 3: Measurement error analysis

Use $|\Delta f| \approx |f_x| |\Delta x| + |f_y| |\Delta y|$.

- (a) ★ Rectangle area $S = ab$, $a = 6$ cm, $b = 9$ cm, precision ± 1 mm. Find maximum absolute error in S .
- (b) ★★ Hollow cylinder volume $V = \pi h(R^2 - r^2)$, $R = 5$ cm, $r = 4$ cm, $h = 7$ cm, precision ± 0.8 mm on radii and ± 2 mm on h .
- (c) ★★ The angle α in a right triangle is computed as $\alpha = \arctan(b/a)$ where $a = 3$ m and $b = 4$ m, each measured with error ± 0.01 m. Find the maximum absolute error in α .
- (d) ★★★ Rectangular prism $V = a^2b$, $a = 20$ cm, $b = 10$ cm, precision ± 4 mm and ± 2 mm. Find the maximum *relative* error $|\Delta V|/V$.

Exercise 4: Chain rule — find the indicated derivatives

- (a) ★ $z = u^2 \ln v$, $u = y/x$, $v = x^2 + y^2$. Find $\partial z/\partial x$ and $\partial z/\partial y$.
- (b) ★ $u = \ln(x + y + z)$, $x = \cos^2 t$, $y = 2t \sin t$, $z = t$. Find du/dt .
- (c) ★★ $z = 2^{x-3y}$, $x = s^2t$, $y = st^2$. Find $\partial z/\partial s$ and $\partial z/\partial t$.
- (d) ★★ $w = e^{xy} + x$, $x = u + v$, $y = e^{u+v}$. Find $\partial w/\partial u$ and $\partial w/\partial v$.
- (e) ★★ $F(x, y) = f(x^2 - y^2, e^{xy})$, f differentiable. Find $\partial F/\partial x$ and $\partial F/\partial y$.
- (f) ★★ $u = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$. Express $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2$ in terms of derivatives with respect to r and θ .
- (g) ★★★ $z = f(x + ay)$, f differentiable. Show that $\frac{\partial z}{\partial y} = a \frac{\partial z}{\partial x}$.
- (h) ★★★ $z = y \varphi(x^2 - y^2)$, φ differentiable. Verify:

$$\frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = \frac{z}{y^2}.$$

Exercise 5: Implicit differentiation

Use $\partial z/\partial x = -F_x/F_z$, $\partial z/\partial y = -F_y/F_z$.

- (a) ★ $x^2 + y^2 + z^2 = 1$. Find $\partial z/\partial x$ and $\partial z/\partial y$.
- (b) ★ $x^2 + y^2 - z = 0$. Find $\partial z/\partial x$ and $\partial z/\partial y$.
- (c) ★★ $e^{\sin z} = z^2 xy$. Find $\partial z/\partial x$ and $\partial z/\partial y$.
- (d) ★★ $xy + yz + xz = 1$. Find $\partial z/\partial x$ and $\partial z/\partial y$.

(e) ★★ $\Phi(cx - az, cy - bz) = 0$ defines z implicitly. Show:

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = c.$$

(f) ★★★ $z \ln(x + z) - xy/z = 0$. Find $\partial z / \partial x$ and $\partial z / \partial y$.

(g) ★★★ If $F(x - z, y - z) = 0$ defines $z = z(x, y)$ implicitly, show that $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1$.

★ *Standard formula* ★★ *Chain rule with two parameters* ★★★ *Abstract identity or implicit differentiation.*